

RVTools to IBM Cloud VPC

Comprehensive Right-Sizing & Resilience Logic Guide

1. The "Why" Behind the Logic

Standard migration tools often look only at **Allocation** (how much RAM/CPU is assigned). This tool focuses on **Utilization and Contention** (how the VM actually performs). Our goal is to ensure Day 1 stability in IBM Cloud VPC by identifying "stressed" workloads that cannot afford to be right-sized.

2. Performance Safety Overrides: Understanding Contention

Before the tool suggests a smaller (cheaper) profile, it checks the source VM for signs of resource contention using two "red flag" metrics:

A. CPU Ready (%RDY) — "The Waiting Room"

- **Definition:** Measures the time a VM is ready to run but is forced to wait because physical CPU resources are busy.
- **Threshold:** > 5.0%
- **The Logic:** If a VM spends > 5% of its time waiting, it is already resource-constrained. Right-sizing this VM would likely lead to performance outages in the cloud.

B. CPU Co-Stop (%CSTP) — "The Sync Delay"

- **Definition:** Measures time multi-vCPU VMs spend waiting to synchronize so they can execute instructions simultaneously on physical cores.
- **Threshold:** > 3.0%
- **The Logic:** High Co-Stop indicates architectural inefficiency (the VM is "tripping over itself"). These must be matched exactly (Safety Match) to ensure stability during the transition.

3. Decision Pillar 2: The N+1 Resilience Math (Availability Parity)

While IBM handles physical hardware failover in VPC, this tool calculates **Logical N+1** to ensure the target architecture matches existing Service Level Agreements (SLAs).

The Calculation

- 1. Total Cluster Capacity (MHz).
- 2. Minus the capacity of the Largest Host in the source cluster.
- 3. Compare the remaining capacity against the Current Total Demand.

Why this matters: For Dedicated Hosts, this ensures that if one host is down for maintenance, the remaining hosts have the headroom to support the evacuated VSIs.

4. Decision Pillar 3: Zombie VM Detection

To optimize cloud spend, the engine identifies "Ghost" workloads consuming resources without providing value.

- **Criteria:** Power State: Powered On | CPU Utilization: < 5% | Total Demand: < 100 MHz.
- **Action:** Flagged for decommissioning to prevent unnecessary migration costs.

5. Decision Pillar 4: Storage Performance & IOPS Tiering

The engine utilizes a "Keyword + Performance" hybrid logic to assign the correct storage profile.

A. Intent-Based Keyword Triggers

Detection of these keywords in the VM Name or Environment automatically triggers a **10 IOPS/GB** tier:

- **Environment Tags:** prod, production, crit.
- **Workload Indicators:** sql, db, oracle, sap, hana.

B. Performance-Based Triggers

- **Disk Latency:** If source telemetry shows sustained disk latency > **15ms**, the engine upgrades the volume to 10 IOPS/GB.

- **Contention Alignment:** VMs flagged for a "Safety Match" (high CPU Ready or Co-Stop) default to 10 IOPS/GB to ensure maximum performance headroom.